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Heat transfer and mixing in initiated chemical vapor deposition analyzed by *in situ* gas composition sensing

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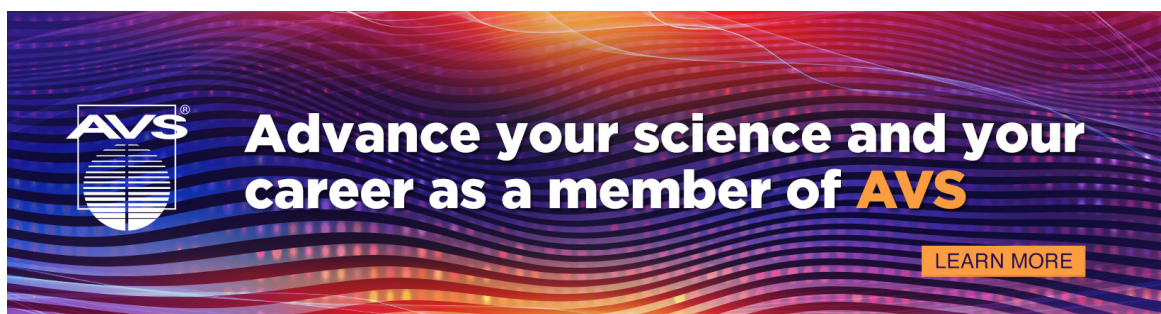
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ABSTRACT

When interpreting or modeling initiated chemical vapor deposition (iCVD) data, assumptions about vapor phase heat and mass transport are almost always required. However, empirically verifying such assumptions is difficult and rarely attempted. To address this gap, we utilize *in situ* gas composition measurements and heat transfer experiments to validate theoretical and computational models of the iCVD system. Gas composition is obtained using a Pirani pressure gauge, which we demonstrate to be effective in simple gas mixtures. Dimensional analysis shows that the parameter space describing heat and mass transport can be simplified significantly, with the degree of mixing in the reactor determined by the Peclet number (Pe) and the heat transfer regime determined by the Knudsen number (Kn). Multiphysics simulations indicate that the well-mixed assumption holds below $Pe = 0.3$. Residence time distribution measurements for argon and isopropanol support this conclusion. Next, we study heat transfer by measuring thermal diffusion from the filament at a range of pressures in argon and isopropanol. These data show that heat transfer is pressure independent above $Kn = 0.4$ and validates a first-principles model of heat transfer over the full range of Kn. To account for thermally driven pressure changes in batch mode, we define an “effective temperature” for the reactor and validate a model of it against pressure versus filament temperature data. Our findings provide a systematic framework for iCVD practitioners to estimate mixing and temperature profiles and assess the impact of design choices on both batch and continuous operation.

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I. INTRODUCTION

Polymer thin films are used widely in applications such as organic electronics¹ and membranes.² Solution phase techniques like spin coating and dip coating are the most popular ways to fabricate polymer thin films; however, the use of polymer chemical vapor deposition (CVD) processes³ broadens the range of applications. Polymer CVD is solvent-free, enables conformal film synthesis,^{4–8} is impervious to surface tension-induced defects like dewetting, and enables precise thickness control even for ultrathin films (<100 nm).⁹ Many polymer CVD processes require harsh conditions (like plasma-enhanced CVD)^{10,11} or have a limited library of polymers that can be synthesized (like parylene CVD).¹² Initiated chemical vapor deposition (iCVD) overcomes these limitations by decoupling initiation from the polymerization reactions. The initiator decomposes at a heated filament, while polymerization occurs

on a cooled stage. As a result, iCVD operates under benign conditions and is compatible with a broad range of chemical functionalities and substrates.¹³

During iCVD, physical processes like heat transfer from the array and chemical processes like thermal decomposition of the initiator occur simultaneously. In complex systems such as this, insights are often gained by using Multiphysics simulations to solve the governing partial differential equations for heat, momentum, and mass transfer. However, we are aware of only two such Multiphysics simulations of iCVD in the literature.^{14,15} Both simulations target novel reactor configurations instead of studying commonly used conditions and, as a result, have not been incorporated into a broader understanding of the iCVD system. Given the importance of Multiphysics simulations to inorganic CVD research,¹⁶ the comparative lack of computational study in iCVD highlights a gap in our understanding.

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Though there are few theoretical or computational studies of iCVD to draw from, elucidation of fundamental processes remains necessary for material design, innovations in synthesis, and optimization for monomer yield¹⁷ and polymer quality. So far, this fundamental understanding comes primarily from studies that describe how deposition rate and final film properties (like molecular weight) are effected by process variables like the reactor pressure and filament temperature.¹³ Because these studies do not measure composition in the vapor phase, efforts to unravel the effects of the vapor phase on deposition kinetics have been sparse. While there is an emphasis on understanding the initiator decomposition and surface polymerization reactions occurring in iCVD, these reactions are almost always affected by mixing and heat transfer.¹⁸ Because mixing and heat transfer can be studied in isolation, a strong model of them is needed to set the foundation for studying reactive processes. While rare, existing efforts to directly measure heat transfer and mixing in the iCVD system warrant discussion.

Regarding heat transfer in iCVD, an excellent experimental investigation of the iCVD filament was performed by Bakker *et al.*, precisely modeling the important radiative effects in heat transfer under variable pressure for vapors relevant to iCVD.^{14,19} The model of thermal diffusion proposed by this study empirically relates filament temperature to heat transfer for the reactor studied and does not explain how system geometry or gas properties affect heat transfer. While the kinetic theory of gases is expected to explain these phenomena,²⁰ there have been no efforts to validate a first-principles theory of thermal diffusion in the iCVD reactor.

In the study of mixing, the primary goal is to determine the contacting pattern in the reactor, which generally involves comparing a measured residence time distribution (RTD) with an ideal well-mixed or plug-flow model.²¹ The only RTD measurement we are aware of in an iCVD reactor was obtained using the response of vapor composition at the outlet of the reactor to gas pulses measured using an *in situ* FTIR.²² While the authors describe their system as a plug flow, the absence of mathematical analysis and the incompatibility of plug flow with good coating uniformity demonstrated by iCVD under similar conditions cast doubt on the broad applicability of this conclusion.³ Indeed, the known dependence of mixing on system variables like flowrate²³ suggests that describing flow as either plug flow or well-mixed flow is incomplete. Instead, dimensionless numbers that correlate with the dominant physical processes driving mixing can capture that dependence quantitatively. While dimensionless numbers have been used in the iCVD reactor scaleup,²⁴ there have been limited efforts to leverage them in the quantitative study of regime changes in iCVD, like the onset of poor mixing or the onset of pressure-dependent heat transfer.²⁵

In iCVD, manipulating well-understood regime changes has been a remarkably fruitful basis for innovation. For instance, deposition at or above the monomer saturation pressure has yielded new insights into polymerization kinetics,²⁶ nanoparticle fabrication,^{27,28} and porous film fabrication.²⁹ We should expect that less well-understood regime changes in heat and mass transfer might enable similar innovations. For instance, poor precursor mixing in the vapor phase is well known to cause film nonuniformity.^{23,24} A formalism to quantify mixing could be used to mitigate the need for batch iCVD^{17,23,30} or be leveraged to produce gradient films (as has been done in parylene CVD).³¹ In the realm of heat

transfer, it has been shown that heat transfer from the filament is strongly affected by the reactor pressure, but there are no methods for predicting the onset of this effect.¹⁹ The ability to do so might enable the use of initiating systems that are only effective at low pressure.

In this paper, we focus on developing models for both mixing and pressure-dependent heat transfer. The end goal is to produce methods that allow researchers to predict the temperature distribution and approximate mixing profile for their reactor using flow rate, pressure, and reactor dimensions. Currently, there is no industry standard method for measuring the temperature of the iCVD vapor, and composition is generally estimated from the vapor flow rate, which is rarely validated by direct measurement. To understand mixing in iCVD, precise, real-time knowledge of the vapor phase composition is key. However, vapor phase composition monitoring via *in situ* FTIR²² and *in situ* mass spectrometry³² is expensive and has not gained widespread usage. Furthermore, they often need to be placed at the outlet of the reactor, potentially introducing a time delay in the measurement. Just as *in situ* interferometry enables precise thickness control, an affordable, precise, and simple-to-implement composition sensor enables low-cost composition-based feedback control during deposition. The work of Schröder *et al.*³² illustrates the benefits of *in situ* gas composition monitoring to ultrathin and copolymer films and further discusses its industrial relevance in detail.

To address the absence of low-cost, precise, and simple real-time vapor phase composition sensing in iCVD, we introduce a method for detecting composition using a Pirani convection gauge. The technique of using the Pirani gauge to obtain real-time vapor-phase composition enables us to compare the RTD obtained from simulations to experiments. Pirani gauges measure pressure via heat transferred from a fine wire. At low pressures, the rate at which molecules collide with the wire limits heat transfer, so the rate of heat transfer is proportional to the pressure.³³ Because larger molecules transfer more energy per collision than smaller molecules, the pressure reported by a Pirani gauge depends on the composition of the vapor. In fact, since there are relatively few collisions between gas molecules in a Pirani gauge, the pressure reported by the Pirani gauge often varies linearly with the mole fraction,^{34,35} making it an especially good composition sensor. We argue that the potential impacts of Pirani gauge-based composition sensing in iCVD extend beyond what we demonstrate here, pointing to benefits from precise process control and facile deployment in research and the thin film industry. For example, composition sensing can improve film quality and consistency by verifying the films are being deposited under the correct target conditions.

We begin our investigation by estimating key dimensionless numbers in iCVD to determine whether the process can be described by an ideal model of heat transfer and mixing, characterized by pressure-independent filament heat transfer and a well-mixed reactor. We show how deviation from the ideal model can be quantified using the Peclet number (Pe_m) (mixing) and Knudsen number (Kn) (heat transfer) via pressure-dependent heat transfer experiments and mixing simulations. While we used argon and isopropyl alcohol (IPA) as model gas species, we demonstrate how our methodology can be generalized to other gases/vapors and reactor geometries. To characterize thermal diffusion, we establish

first-principles models that can be generalized to most other reactor geometries to provide predictions of the temperature profile in the reactor. While mixing is more complex, we show that simple, low-cost Multiphysics simulations can be used to evaluate the effect of design choices and estimate the composition throughout the reactor. We hope our work will provide iCVD researchers with the tools to obtain accurate estimates of the mixing state of their reactor, heat transfer in their reactor (including the temperature profile), and how both these phenomena are affected by flow rates, gas composition, pressure, reactor dimensions, and reactor temperatures.

II. EXPERIMENT

A. Mixing simulations

Reactor models used in Multiphysics simulations were built in Autodesk Inventor 2025 (Autodesk, San Francisco, California) to match a standard³⁶ two-chamber iCVD reactor used in experiments and a one-chamber alternative. The two-chamber design has a top chamber height of 2.25", a bottom chamber height of 3", and a diameter of 9.625". The one-chamber design has a height of 2.5" and a diameter of 9.625". All dimensions represent dimensions of the vapor volume, i.e., internal dimensions. Schematics for these designs are shown in Fig. 1.

Mixing simulations were performed in Autodesk CFD 2024 (Autodesk, San Francisco, California). A computational mixing study was performed with two gases flowing into the reactor through two separate inlets at a respective nominal flow rate of 1 SCCM at steady state. The composition at the reactor stage was reported for diffusion coefficients of 4, 16, and 64 cm² s⁻¹, corresponding to Pe_m of 5.3, 1.3, and 0.3, respectively at the simulation reactor pressure of 0.1 Torr and 298 K. These conditions were chosen to span the upper range of Pe_m for the iCVD system and illustrate the onset of poor mixing.

Transient step-response experiments (Sec. II D) were simulated for the two-chamber reactor under a constant total flow of 1 SCCM. To induce the step response, first, a steady, fully developed 1 SCCM flow of a primary gas through one inlet was established, and then simultaneously, a 1 SCCM flow of a secondary gas through the other inlet was initiated while the flow of the primary gas was ceased. Step-response curves showing the change in composition were reported at three locations: the outlet of the reactor, near the inlet in the top chamber (side port), and far from the inlet in the bottom chamber (back port). Simulations were performed with diffusion coefficients of 4, 16, 64, and 678.5 cm² s⁻¹, corresponding to Pe_m of 5.3, 1.3, 0.3, and 0.03 at the simulation reactor pressure of 0.1 Torr and 298.15 K. The simulation at Pe_m of 0.03 was included to match the conditions for the RTD of IPA and argon measured in Sec. II D. Videos of the changing composition are included as [supplementary material](#) and follow the naming convention "X_cm2pers.AVI," where X is the diffusion coefficient used.

B. Flow calibration

Argon gas, obtained from Airgas (Radnor, Pennsylvania), and 99.7% IPA obtained from Sigma Aldrich (Burlington,

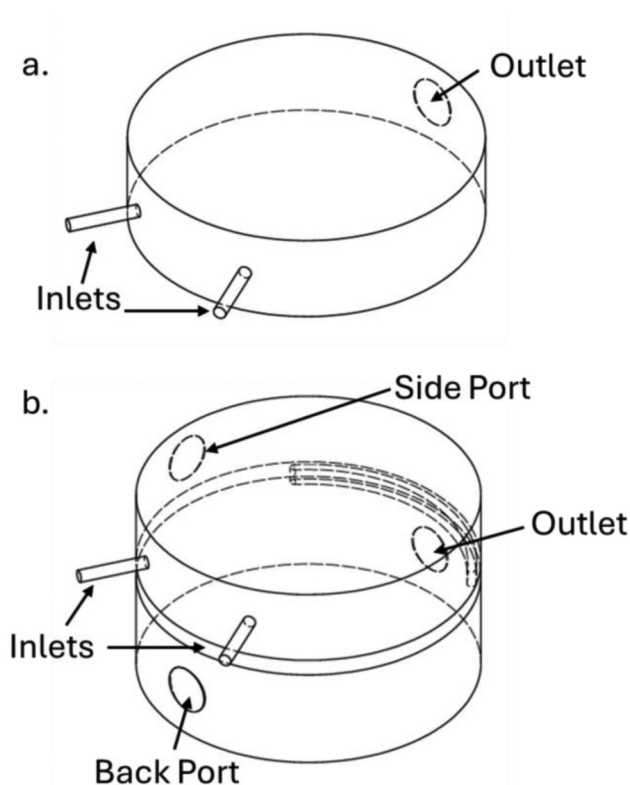


FIG. 1. Schematic of the two iCVD reactor geometries used in simulations; (a) single-chamber design with inlets and outlet labeled and (b) double-chamber design with inlets, outlet, side port, and back port labeled. This double-chamber design matches the design used in experiments.

Massachusetts) were used as model gases in all experiments and were both used without further purification. Flow of argon gas was regulated by a mass flow controller type 1159B from MKS instruments (Rochester, New York), while the flow of IPA was controlled using an SS-4BMW-VCR needle valve from Swagelok (West Henrietta, New York).

Flow rate measurements used in all experiments for IPA and argon were obtained by flow calibration. First, the leak rate of the reactor was measured by closing the outlet to the pump at base pressure (2–5 mTorr). A gradual increase in pressure due to leak was then monitored using an MKS Baratron capacitance manometer model 626C from MKS instruments (Rochester, New York) by recording pressure every 10 s for 2 min and performing a linear regression. The slope of the pressure versus time plot was converted to a flow rate in standard cubic centimeters per second (SCCM) using the effective temperature (see Sec. III E). The flow rates of gases were measured by setting a flow condition using either the mass flow controller (for argon) or the needle valve (for IPA). The outlet to the reactor was then closed, and the increase in pressure was measured every 2 s for 20 s. A linear regression was performed on the pressure versus time plot, and the slope was converted to a

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molar flow rate in SCCM using the ideal gas law. Three replicates were performed for each flow rate.

C. Vapor phase composition measurements using a Pirani gauge

For binary gas mixtures in the iCVD reactor, the composition-sensitive Pirani gauge pressure is sufficient to deduce composition with good accuracy. First, a standard curve is created by measuring both the Pirani and Baratron pressures at steady state, while varying the vapor-phase composition by changing the flow rates of each of the mixing gases (keeping the total flow rate constant). The ratio of the Pirani gauge pressure to the Baratron pressure is then plotted against the vapor-phase composition to create a standard curve (generally linear). In scenarios where the reactor composition is unknown, but the Pirani gauge pressure and Baratron pressures can be read, the standard curve can be used to determine the vapor-phase composition. The Pirani gauge was attached to the side port of the reactor as labeled in Fig. 1(b).

D. Residence time distribution measurement

Step-response curves were generated by flowing argon into the reactor at a fixed nominal flow rate of 1 SCCM and then rapidly shutting off the argon flow and opening IPA flow into the reactor such that the total flow rate remained constant. This caused a step change in the composition of inlet vapor, which was registered by a Pirani gauge at the side port. Flow through the IPA needle valve is choked, so the flow rate of IPA depends only on the upstream pressure, which is expected to remain constant throughout the experiment. While this suggests that the IPA flow reaches steady state instantaneously, there are significant pressure shocks caused by the actuation of ball valves that were corrected for in data analysis. Composition at the side port was determined using a standard curve (Sec. II C) matched to IPA/argon mixtures.

E. Effective temperature measurement

In all experiments, an 80/20 Ni Cr nichrome wire (Omega Engineering, Stamford Connecticut) was used for the filament array. The following procedures were used to collect data used to determine the effective temperature in the reactor in Sec. III E. These procedures were performed for nominal chamber pressures of 50, 100, 200, and 300 mTorr for both pure argon gas and pure IPA vapor. First, the reactor was pressurized to the nominal chamber pressure. Next, with no power being delivered to the array, the pressure, filament temperature, and stage temperature were logged every 5 s for 60 s. Initial leak rate was calculated from the rate of pressure change during this period (detailed in Sec. II B). Initial filament temperature was taken as the average filament temperature over the first 60 s period. Subsequently, power to the filament array was stepped up by ~ 10 W every 60 s until the maximum power, i.e., ~ 80 W, was reached. Over the course of each 60 s period where power is constant, the pressure, filament temperature, and stage temperature were logged every 5 s.

F. Heat transfer in argon and IPA vapor

Heat transfer from the filament array to the reactor body can be determined by performing an energy balance on the filament array adapted from Bakker *et al.*¹⁹ Heat is generated resistively in the filament and is radiated and conducted away from the filament. At low pressures, radiative heat transfer dominates and a radiation-based model that depends only on the filament temperature and stage temperature can be used. At higher pressures, the radiative heat transfer calculated from this model is subtracted from the total dissipated energy to determine the heat transfer by conduction to the gas.

Heat transfer measurements are performed in pure argon gas and pure IPA vapor flowing through the reactor at nominal pressures of 25, 50, 75, 100, 150, 200, and 1000 mTorr and filament temperatures ranging from ~ 50 to ~ 350 °C and at a constant 1 SCCM flow rate of either argon gas or IPA vapor (Sec. II B). Power to the filament was measured by multiplying the voltage and current from a BK precision DC power supply (model 1710A, Yorba Linda, California). Temperature of the filament array was measured using a K-type thermocouple spot-welded to the filament array so that the thermocouple junction was less than 2 mm from the filament to ensure good thermal contact. To verify that the current passing through the filament array did not impact the temperature measured by the thermocouple, the current was reversed, and the temperature measurement did not change. Measured values of the filament wire length (2.7 m) and filament diameter (0.4 mm) were used in calculations in Sec. III E. Length was determined using total resistance across the array and the known resistance of the wire at 20 °C of $8.5 \Omega \text{ m}^{-1}$, while diameter was determined using an instant readout digital caliper from Electron Microscopy Sciences (Hatfield, Pennsylvania).

III. RESULTS AND DISCUSSION

A. Dimensional analysis

The first step in modeling heat transfer and mixing in iCVD is to determine which physical phenomena (diffusion, convection, radiation, etc.) most significantly impact the state of the reactor. The resulting assumptions simplify both the development and implementation of models. Dimensionless numbers are a well-established method to assess the relative importance of physical phenomena and establish the governing equations for a system. Because the discussion of dimensionless quantities in iCVD literature has been limited to reactor scaleup,²⁴ further analysis is needed to develop accurate models of heat transfer and mixing.

Dimensionless numbers relevant to mass transfer in the iCVD system are the Reynolds number (Re), Mach number (Ma), Peclet number for mass transport (Pe_m), and Knudsen number (Kn) (summarized in Table I).²⁴

Dimensionless numbers relevant to the transfer of heat include the Peclet number for thermal transport (Pe_t), the Grashof number (Gr), and the Knudsen number (summarized in Table II). We do not include dimensionless numbers relevant to the absorption of infrared radiation emitted by the filament because estimating the molar absorptivity of gases is nontrivial and depends on both the emission spectrum of the filament and the absorbance

TABLE I. Dimensionless numbers relevant to mass transfer in iCVD.

Name	Equation	Description	Expected range	Transitional range
Peclet (mass)	$Pe_m = \frac{lv}{D}$	Advective versus diffusive mass transport	10^{-6} – 10	10^{-1} – 10
Knudsen	$Kn = \frac{\lambda}{l}$	Length scale of molecular collision to the scale of the system	10^{-5} – 10^{-1}	10^{-1} – 10 (Ref. 37)
Reynolds	$Re = \frac{\rho v l}{\mu}$	Inertial versus viscous forces	10^{-8} – 1	2000–3000 (Ref. 38)
Mach	$Ma = \frac{v}{c_s}$	Characteristic velocity to the speed of sound	10^{-5} – 10^{-2}	>0.3 (Ref. 39)

spectrum of the gas. To address the issue of absorbance, we instead compare experiments for absorbing and nonabsorbing gases in Sec. III D.

Transitional ranges are based on external reference,²⁵ or in the cases of Kn for heat transfer and Pe_m for mass transfer, our own data are presented in further sections. The dimensionless numbers in Tables I and II (rounded to the nearest power of 10) were estimated from ranges detailed below, which are based on commonly used research conditions.³

We have assumed a typical lab-scale reactor with characteristic length scale (l_{body}) ranging from 1 cm (for tubing connections) to 30 cm (long reactor dimension) and a filament radius (R_{fil}) of 0.20 ± 0.02 mm. The length scale used to calculate Gr was a typical separation distance between the stage and the filament of 3 cm, which does not vary significantly in the literature.³ The range of reactor conditions is based on common deposition conditions,¹³ with stage temperatures (T_{stage}) between 0 and 50 °C, filament temperature (T_{fil}) from 150 to 400 °C, vapor flow rates from 0.1 to 1 SCCM, and chamber pressure from 50 to 1000 mTorr. Vapor flow rates and the ideal gas law were used to determine characteristic velocity (v) using the cross-sectional area of a typical lab-scale reactor with an approximate height of 5 cm and a long-reactor dimension (width or diameter) of 30 cm. In the calculation of Gr , the acceleration due to gravity was 9.81 m s^{-2} .

Gas density (ρ) and heat capacity (c_p) were determined for ideal gases with three rotational and three translational degrees of freedom (heat capacity ratio, γ , of 4/3). Diffusivity (D), dynamic viscosity (μ), mean free path (λ), and speed of sound (c_s) were calculated based on correlations found in Poling⁴⁰ and Lide⁴¹ for either a “large” or “small” process gas molecules. The following gas properties used in estimations were based on tabulated data for molecules similar to those used in iCVD.^{40,41} The thermal conductivity (κ) was estimated directly from tabulated data⁴² and the assumption that $\kappa \sim \sqrt{T}$ from kinetic theory of gases⁴¹ to range from $0.001 \text{ T}^{1/2}$ to $0.0005 \text{ T}^{1/2} \text{ W m}^{-1} \text{ K}^{-1}$ for “small” and “large”

cases, respectively. Molecular weight (M) of process gases was assumed to range from 50 to 500 g mol^{-1} ,¹³ boiling point of process gases from 200 to 50 °C (based on values reported by suppliers for common monomers), and critical molar volume (V_c) of process gases between 200 and $1000 \text{ cm}^3 \text{ mol}^{-1}$ for “small” and “large” cases, respectively.⁴¹ The temperature (T) used to determine gas properties was assumed to range from the minimum stage temperature (0 °C) to the maximum filament temperature (400 °C).

For the calculation of mean free path, viscosity, and diffusivity, the collision diameter (d) of the process gases was calculated using Eq. (9-48) from Poling.⁴⁰ In this correlation, the method of Tyn and Calus was used to calculate the molar volume of process gases at their boiling point as reported in Lide.⁴¹ The mean free path was calculated from kinetic theory using Eq. (1),⁴¹

$$\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 p}, \quad (1)$$

where p is the chamber pressure and k_B is Boltzmann’s constant. Diffusivity and dynamic viscosity of process gases were calculated using the method of Wilke and Lee as reported in Poling.⁴⁰ The speed of sound was calculated using the relationship for ideal gases given by Eq. (2),⁴³

$$c_s = \sqrt{\frac{\gamma R_G T}{M}}, \quad (2)$$

where R_G is the ideal gas constant.

Table I summarizes dimensionless numbers relevant to mass transfer in iCVD. Re is the ratio of inertial to viscous forces in a flow and is often used to establish the onset of turbulence.³⁸ Because vapors in iCVD are low density, viscous stresses dominate inertial stresses under typical conditions, so Re is small, and flow is laminar.²⁴ Ma relates the speed of sound to the characteristic

TABLE II. Table of dimensionless numbers relevant to heat transfer in the iCVD system.

Name	Equation	Description	Expected range		Transitional range
			$l = R_{fil}$	$l = l_{body}$	
Grashof	$Gr = \frac{g \rho^2 (T_{stage} - 1) \beta}{\mu^2}$	Buoyant versus viscous force	10^{-5} – 10^{-1}		$>10^2$
Péclet (thermal)	$Pe_t = \frac{lv \rho c_p}{\kappa}$	Advection vs conduction	10^{-8} – 10^{-3}	10^{-6} – 1	10^{-1} – 10
Knudsen	$Kn = \frac{\lambda}{l}$	Scale of molecular collisions versus system scale	10^{-2} – 10	10^{-5} – 10^{-1}	$>10^{-1}$

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velocity of the flow and is often used to determine whether a flow is incompressible.³⁹ The typical range of volumetric flow rates used in iCVD (discussed above) results in low characteristic velocities, meaning Ma is much less than its transitional value³⁹ of 0.3, and compressibility effects can be neglected. Because $Ma \ll 0.3$ and $Re \ll 2000$,³⁸ we determined that the typical vapor flow in iCVD is laminar and incompressible. Furthermore, because $Re \ll 1$ in most cases, the flow profile in the reactor is well-described by equations for the Stokes flow.³⁸

Both Pe_m and Kn are of a transitional scale, meaning that the effects they describe can only be neglected sometimes. Kn is the ratio of the mean free path to the length scale of the system. When Kn is less than ~ 0.1 , the flow is “viscous” and can be described by continuum fluid mechanics.³⁷ Kn is anticipated to be of transitional scale in confined parts of the reactor (such as tube connections) when small process gases are used at low pressures. Under these conditions, flow transitions to Knudsen flow, a regime between viscous and molecular flow, where continuum assumptions break down.³⁷ Because our study focuses on the bulk reactor and Knudsen flows primarily occur in tube connections, we will not investigate Knudsen flows further. It should be noted that Knudsen flow can be important for reactor design since it is why long, small-diameter tubing connections increase pump-down times.

Pe_m describes the ratio of advection to diffusion. When Pe_m is small (for low flow rates or high diffusivity mixtures), the well-mixed assumption is valid. Because well-mixed conditions correspond to uniform coating in iCVD,³⁰ the well-mixed condition is the ideal mixing condition. As Pe_m increases, its value quantifies the departure from perfect mixing. Because the expected range of Pe_m for the iCVD system overlaps with the transitional range²⁵ of Pe_m (Table I), we will investigate it further in Secs. III B and III C. Because the mixing condition of the reactor is determined almost solely by Pe_m , the mixing condition for any vapor mixing can be determined if flow rates and diffusivity are known or can be estimated.

Table II summarizes dimensionless numbers relevant to heat transfer in iCVD. It should be noted that we do not include dimensionless numbers related to radiation in this discussion. While thermal radiation makes up a significant fraction of the heat emitted by the filament,¹⁹ our experiments in Secs. III D and III E provide no evidence that radiation impacts process gases except by heating the reactor walls. Pe_t for heat transfer from the filament is small, and heat transfer is predominantly conductive because heat flux is inversely proportional to distance from a slender body like the heated filament in iCVD,⁴⁴ and most of the conductive resistance to heat transfer occurs close to the filament surface (see Sec. III D for a more detailed discussion). Because Pe_t is small on the scale of the filament array (when the characteristic length scale is the filament radius), the role of thermal advection in heat transfer is expected to be small. We will leave any investigation of thermal advection in iCVD for future work.

Kn is defined the same for heat transfer as it is for mass transfer. In heat transfer, our primary interest in Kn is over the length scale of the filament array. Kn based on filament length scales describes the effect of low pressure on heat transfer near the filament. On reactor length scales, Kn is small enough that its effects can be ignored.

Gr describes the ratio of buoyant to viscous forces in a fluid. When Gr is above 10^2 , natural convection begins to occur. Though there has been speculation about the role of natural convection in iCVD,¹⁵ for the typical conditions in Table II, we find that the Grashof number is three orders of magnitude below the range at which natural convection is known to occur. As such, the description of heat transfer from the walls of the reactor can be described primarily by Fourier conduction, with some pressure dependence at low Kn . Our discussion of dimensionless numbers in heat transfer does not seek to be fully comprehensive over all potential iCVD conditions but accounts for the most significant impacts on heat transfer to the main process gas for typical operating conditions.

B. Mixing simulations

Simulations were performed to help visualize the mixing between two distinct gas flows entering the reactor through separate inlets. Two reactor geometries were used. The double-chamber geometry matched the reactor used in experiments, and a single-chamber equivalent was simulated to show the effects of different chamber designs. While two-chamber designs are common, the bottom chamber acts as a difficult-to-model dead zone. Steady-state mixing simulations were performed with a constant flow of gas into the reactor through both inlets at 1 SCCM.

To obtain models of the full range of Pe_m , we varied the diffusion coefficient. Simulations were performed for Pe_m of 0.3, 1.3, and 5.3 in both the one-chamber and two-chamber designs (Fig. 2), representing transitional Pe_m of roughly 0.1–10. Pe_m for IPA/argon mixing system was calculated based on the binary diffusion coefficient of IPA and argon reported in Lee,⁴⁵ scaled down to a pressure of 100 mTorr by assuming diffusivity is inversely proportional to pressure based on kinetic theory of gases.⁴¹ We have calculated Pe_m from the individual gas flow rates, not the overall gas flow rates such that each gas flow is associated with a different value of Pe_m . The simulation results show that at Pe_m below 0.3, the reactor is well-mixed (Fig. 2). Simulations were also performed for Pe_m of 0.03, roughly corresponding to the mixing of argon and IPA studied in Sec. III C. Under Pe_m of 0.03, the system was also well-mixed (details can be found in the supplementary material). Because poor mixing would be associated with depositions that are nonuniform and difficult to repeat or characterize, these simulations indicate that for good performance, iCVD reactors should be operated below Pe_m of 0.3.

It is notable from simulations (Fig. 2) that the increased volume from the two-chamber design has very little effect on the transitional Pe_m ; as such, the appropriate length scale to use in calculating Pe_m is the distance from the inlet to the outlet. The minor difference in the composition profile of the one-chamber versus two-chamber design near the outlet of the reactor is caused by the slit-shaped connection between the top and bottom chambers for the two-chamber design (Fig. 1).

While we only show simulations of circular “pancake” type reactors, if Pe_m for a more exotic reactor geometry needs to be calculated, the longest dimension should be used as the length scale. This is because during actual deposition, not only do inlet streams need to be well mixed, but degradation products of the preinitiator must also be well-mixed (hence, plug flow is also a nonideal mixing

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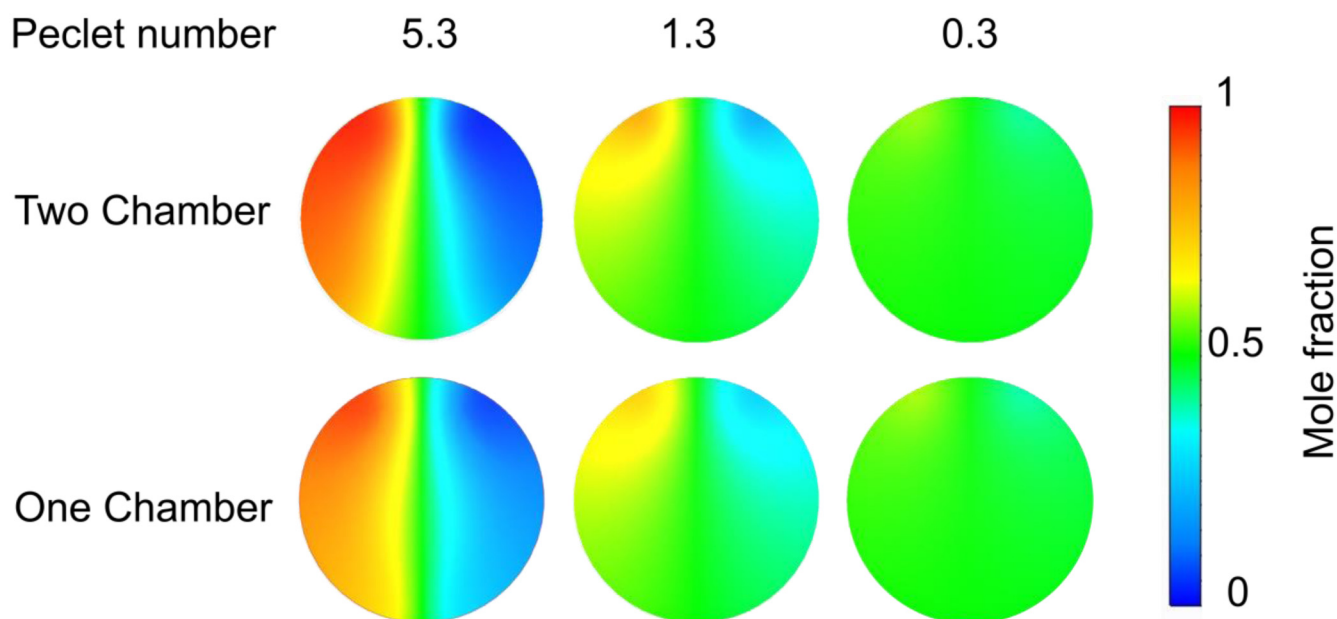


FIG. 2. Binary gas mixing corresponding to the transitional range of Pe_m in two-chamber and one-chamber reactor configurations.

scenario). More information on the choice of characteristic dimensions can be found in textbooks that discuss dimensional analysis and similitude.²⁵ To that end, it should be further noted that while Pe_m describes nonideal mixing, nonuniform depositions can still occur even at low Pe_m when deposition rates are high.²⁹

Transient simulations were performed to provide insight into the step-response experiments performed in Sec. III C. While impulse response curves are generally preferred to step response in determining contacting patterns and mixing,²¹ generating and detecting an impulse without significant pressure-driven flows were infeasible in our setup. Simulations were run to exactly replicate the procedures used in Sec. III C. Figure 3 shows step response for all three Pe_m of 5.3 [Fig. 3(a)], 1.3 [Fig. 3(b)], and 0.3 [Fig. 3(c)]. Simulations were also performed for a Pe_m of 0.03, roughly corresponding to the mixing of argon and IPA studied in Sec. III C, and this curve could not be distinguished from the theoretical ideal. Videos of the simulations can be found in the [supplementary material](#) as well as data for $Pe_m = 0.03$.

The step-response curves in Fig. 3 enable two important conclusions. First, for Pe_m associated with poor mixing in Fig. 2, significant deviation from ideal mixing response was observed via the side port and backport, demonstrating that step-response experiments (Sec. III C) could assess the degree of mixing for a given set of conditions. Second, the data demonstrate clearly the sensitivity to sensor location.

In practice, relying on the shape of the step-response curve can introduce significant errors since generating the step change can induce moderate pressure-driven flow in the system, which is not easy to decouple from time delay at high Pe_m . Furthermore, precise step-response data require accurate flow rate calibration,

which tends to be subject to experimental error when working with low diffusivity gases. Even for an extremely poor mixing scenario [Fig. 3(a)], it may be difficult to confirm poor mixing if the Pirani gauge is placed on the outlet of the reactor, so this should be avoided. Because gas-sensing techniques like in-line FTIR often analyze the exhaust gas from the reactor,²² they are much less valuable for real-time sensing of reactor dynamics. Furthermore, at a Pe_m of 1.3 [Fig. 3(b)], it may become challenging to decouple transient behavior from valve actuation from nonideal mixing behavior, even if sampling at the back port or side port is performed.

The simulations in Fig. 3(c) show that if precise mixing information over the transitional regime is needed, sensors should be placed in multiple locations far apart in the reactor, and their difference or ratio should be used to provide a sensitive measurement of the mixing state. Step-response curves in single-chamber reactor configurations may differ from Fig. 3. However, the correspondence between one-chamber and two-chamber designs in Fig. 2 suggests the insights from Fig. 3 on sensor placement that apply to other reactor geometries.

C. Step-response measurements

In situ composition sensing has been used on rare occasions in the iCVD literature,^{22,32} However, a rigorous study of contacting patterns in iCVD and the phenomena responsible for them has been absent. In contrast to costly custom systems for composition sensing,^{22,32} we present a simple, off-the-shelf method of determining binary composition data using a convection-assisted Pirani pressure gauge.

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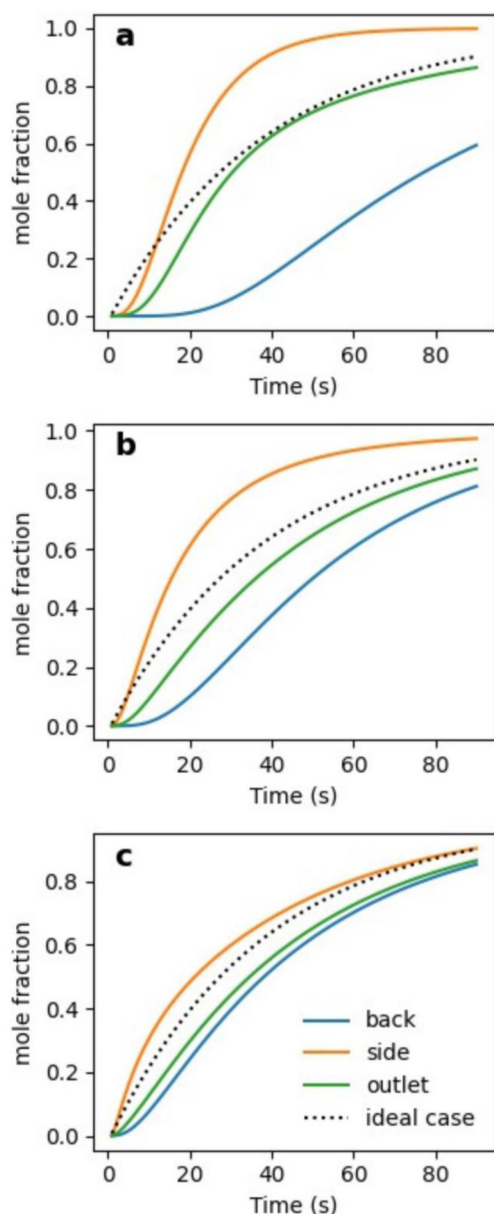


FIG. 3. Step-response curves for two-chamber reactor configuration comparing ideal mixing to various mixing scenarios simulated using different Pe_m . The mole fraction of a gas species introduced as a step function into the chamber is sampled at three locations in the simulation: the back port of the reactor (blue, bottom), the side port (orange, top), and the reactor outlet (green, middle). Simulations are presented for (a) Pe_m of 5.3, (b) Pe_m of 1.3, and (c) Pe_m of 0.3. The ideal case is presented as dotted curves in all cases.

The Pirani gauge uses the pressure dependence of heat transfer to determine pressure and provides a precise, but composition-dependent, measurement of pressure in the range of 1 mTorr to 1 Torr,³⁴ which corresponds to the pressure range for iCVD.¹³

Because heat transfer in low-pressure gases depends on the composition of a gas, Pirani gauges are not appropriate for determining pressure in an iCVD system; however, when combined with a Baratron capacitance manometer, the Pirani gauge becomes an effective composition sensor. Thermal conductivity is routinely used for gas composition sensing,⁴⁶ so the use of the Pirani gauge is not a significant departure from existing methods. However, outside our related work,⁴⁷ it has never been used in the iCVD literature, so our methods represent an exciting new capability for monitoring and controlling the reactor environment. In all experiments, measurement has been performed by a single Pirani gauge attached to the side of the reactor, labeled as “side port” in Fig. 1(b).

To measure the composition, a standard curve is first created [Fig. 4(a)] in which IPA vapor and argon gas are flowed into the iCVD chamber to obtain a range of steady-state compositions. The pressure reported by the Pirani gauge is then divided by the pressure reported by the Baratron capacitance manometer to yield a proxy for composition, which we will refer to as the Pirani gauge ratio (c). The resulting standard curve is linear and can be used to convert the Pirani gauge ratio to gas composition.

Real-time composition sensing from the Pirani gauge enables step-response experiments in which argon is initially flowed at a steady state, and then it is quickly shut off and replaced with a flow of IPA vapor. This experiment maintains a constant total flow rate and pressure. The resulting step response curve [Fig. 4(b)] shows a characteristic well-mixed response (often called a continuous stirred tank reactor or CSTR model).²¹ As such, the well-mixed assumption describes the contacting pattern for the reactor conditions and flow rates studied. The peak of low-residence time material, inconsistent with the well-mixed exponentially decaying RTD [Fig. 4(c)] is indicative of a slight pressure-driven flow when the inlet composition is changed. While a pulsed response is generally more sensitive to nonideal reactor conditions,²¹ we use a step response to maximize the sensing resolution, minimize the effect of pressure-driven flows, and maintain signal integrity. A step change is much easier to introduce at constant flow rate and pressure than a pulse that approximates a delta function in an iCVD reactor.

Based on the measured flow rate of 1.04 SCCM into the reactor, the step-response curve indicates a reactor volume of 5.7 l, which is close to the expected reactor volume measured of 6.3 l (see [supplementary material](#) for calculation) and suggests a well-mixed reactor. Multiple sensors would be required for a more precise study as discussed in [Sec. III B](#). These results are not surprising, given the expectation that the conditions measured represent a Pe_m of 0.03 (based on reactor dimensions and a calculated diffusion coefficient of $678.5 \text{ cm}^2/\text{s}$).⁴⁵ Furthermore, our earlier simulations indicated that the reactor would behave as a well-mixed system at Pe_m of 0.03 (Figs. 2 and 3). Nevertheless, the realization of a well-mixed vapor phase is in marked contrast to the current literature,²² which has concluded that the iCVD reactor system is a plug flow system. We believe the contrast may be due in part to the pulse response curves used in prior work,²² which are associated with the aforementioned pressure-driven flows. Past simulations of iCVD reactors, which show poor mixing, have generally utilized reactor geometries and conditions that differ significantly from standard research conditions.^{14,15}

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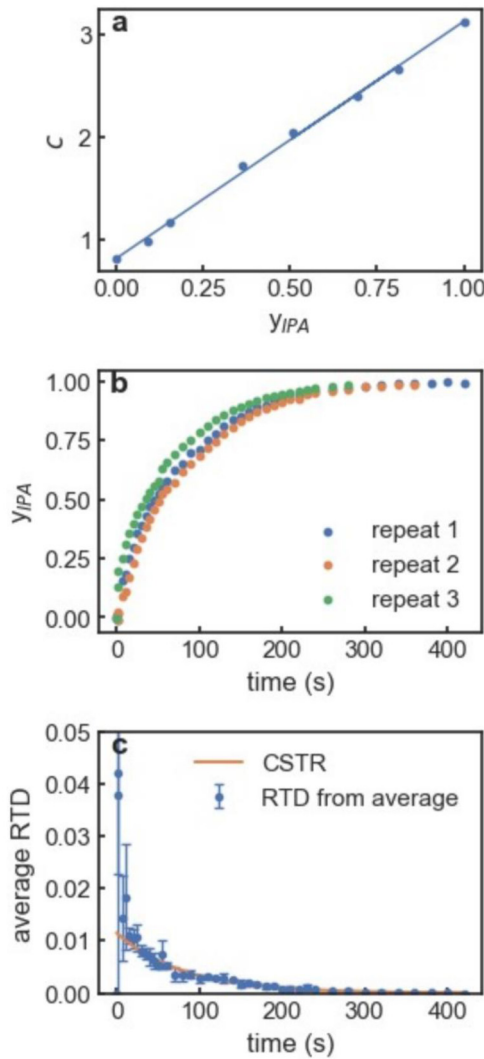


FIG. 4. Measured step-response and RTD for the IPA and argon mixing system. (a) Standard curve correlating the Pirani gauge ratio (c) to the mole fraction of IPA in a vapor phase made of IPA and argon. (b) Experimentally measured step-response curves for the mole fraction of IPA vs time (IPA was introduced at $t = 0$), calculated from the IPA/argon standard curve in (a). (c) The average RTD for IPA, derived from the step-response curves shown in (b), which fits well to a well-mixed ideal model (labeled CSTR).

It is clear from dimensional analysis (Sec. III A) that the iCVD reactor mixing system is determined by Pe_m and Kn , with Re and Ma indicating that the flow is decisively laminar and incompressible. The measured step-response curves for IPA and argon describe a well-mixed system, as predicted by simulation results in Sec. III B. Notably, the binary diffusivity of the IPA and argon system is much higher than normal iCVD conditions, suggesting that nonideal mixing may occur in iCVD when low diffusivity

gases are used. Together with Secs. III A and III B, the data in Fig. 4 support the use of the well-mixed assumption for the iCVD reactor at low Pe_m , with reduced coating thickness/composition uniformity as Pe_m increases.

D. Heat transfer modeling

The work of Bakker *et al.*¹⁹ established the measurement of heat transfer from the filament wires in iCVD, developing a radiation model to describe heat transfer at low pressure. Empirical models for the diffusive heat transfer from the filament were also introduced by Bakker *et al.* However, the derivation of first-principles models of diffusive heat transfer is necessary to predict heat transfer in novel reactor geometries or vapors.

We perform experiments similar to Bakker *et al.* in which we measure filament temperature while varying the pressure in the reactor and the power delivered to the filament array. Following from Bakker *et al.*,¹⁹ heat transfer by conduction from the filament array in $W\ m^{-1}$ (q_{fil}) is given by Eq. (3) in accordance with radiation between gray bodies, when the visible area of one (the walls of the reactor) is much larger than the other (the filament),⁴⁸

$$q_{fil} = \frac{RI^2}{L_{fil}} - 2\pi R_{fil}\sigma\epsilon_{fil}(T_{fil}^4 - T_{stage}^4), \quad (3)$$

where R is the resistance of the nichrome filament wire, I is the current through the filament, L_{fil} is the length of the filament wire, σ is the Stephan Boltzmann constant, R_{fil} is the radius of the filament wire, and ϵ_{fil} is the emissivity of the oxidized nichrome wire, which has a value of 0.95–0.98.⁴⁸

To model diffusive heat transfer in the iCVD vapor, we first consider that diffusive heat flux in $W\ m^{-2}$ (j_q) follows Fourier's law given by Eq. (4),

$$j_q = -\kappa \nabla T, \quad (4)$$

where κ is the thermal conductivity. At steady state, conservation of energy is given by Eq. (5),

$$\nabla \cdot j_q = 0. \quad (5)$$

According to the kinetic theory of gases³³ $\kappa \sim \sqrt{T}$, therefore, we can define a heat transfer potential ϕ from Eq. (4) such that $j_q = \nabla \phi$ to rewrite Eq. (5) as Eq. (6),

$$\nabla^2 \phi = 0, \quad (6)$$

where ϕ is defined by Eq. (7),

$$\phi = -\frac{2}{3}\kappa' T^{\frac{3}{2}}, \quad (7)$$

where κ' is the temperature independent thermal conductivity given by Eq. (8),

$$\kappa' = \frac{\kappa}{\sqrt{T}}. \quad (8)$$

To obtain a model for heat transfer in the reactor from the filament [Fig. 5(a)], we must now solve Eq. (6) for the reactor vapor phase. We will simplify the reactor geometry by ignoring the edges

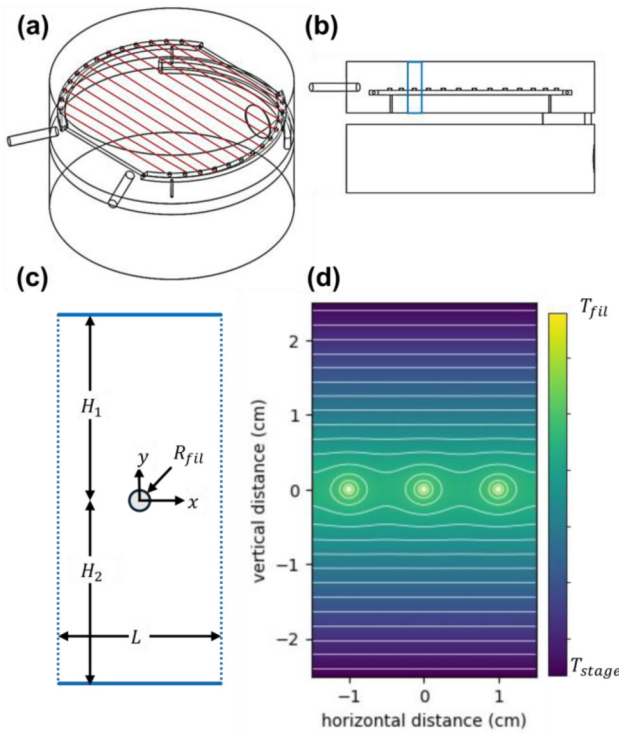


FIG. 5. Schematic of filament array placement and heating. (a) Oblique view of the two-chamber reactor showing the placement of the filament array in the top chamber. (b) Side view of the reactor with the (blue) box corresponding to the schematic in (c). (c) Schematic showing the coordinate system used in the modeling of temperature at the filament array. (d) Plot showing Eq. (9) for three adjacent filament wires as a visual aid.

of the filament array so we can treat it as a 2D problem given by the domain shown in Figs. 5(b) and 5(c), where we have a boundary condition at the stage and lid ($y = -H_2$ and H_1) that $\phi = \phi_{\text{stage}}$ and ϕ_{lid} , respectively, and the periodic boundary condition that $j_q = 0$ at $x = -L/2$ and $L/2$, where x is horizontal distance from the filament; y is vertical distance from the filament; H_1 and H_2 are the distances from the filament to the lid (22.86 ± 0.02 mm) and stage, respectively (31.75 ± 0.02 mm); and L is the distance between filaments (14.7 ± 0.2 mm) as shown in Fig. 5(c). Next, we note that the filament wires are close to each other relative to the height of the chamber, meaning that the solution to Eq. (6) for an infinite line of point sources closely approximates the boundary conditions in Fig. 5(c). This assumption is consistent with the design of CVD reactors with heated filaments because sufficiently large filament spacing to cause nonuniform stage heating also causes nonuniformity in the film.⁴⁹

To find the solution to Eq. (6) for an infinite line of sources in two dimensions, we can draw from the field of complex analysis, in which the Cauchy–Reimann equations show that the real and imaginary parts of any complex function are solutions to Laplace’s equation [Eq. (6)].⁵⁰ Therefore, obtaining

the solution for Eq. (6) for an infinite line of equally spaced 2D point sources (line sources) consists of finding a complex function with an infinite line of equally spaced singularities. Just as $\text{Re}[\ln(z)]$ corresponds to a single-point source in x, y -coordinates when $z = x + iy$, it can be seen that $\text{Re}[\ln(\sin(z))]$ corresponds to an infinite line of point sources (elaboration on this can be found in the [supplementary material](#)). When complex numbers are converted to Cartesian coordinates, and constants of integration are determined from the boundary conditions discussed above, we obtain Eq. (9) (see [supplementary material](#) for details of the derivation),

$$\begin{aligned} \phi = \phi_R &+ \frac{q_{\text{fil}}}{2\pi} \ln \left(\frac{L}{R_{\text{fil}}\pi} \sqrt{\sin^2 \left(\frac{x\pi}{L} \right) \cosh^2 \left(\frac{y\pi}{L} \right) + \cos^2 \left(\frac{x\pi}{L} \right) \sinh^2 \left(\frac{y\pi}{L} \right)} \right) \\ &+ \left(\frac{\phi_{\text{lid}} - \phi_{\text{stage}}}{H_1 + H_2} - \frac{q_{\text{fil}} H_1 - H_2}{2L H_1 + H_2} \right) y, \end{aligned} \quad (9)$$

where ϕ_R is the value of ϕ at the surface of the filament, analogous to T_e in Su *et al.*²⁰ (distinct from the heat transfer potential of the filament itself, ϕ_{fil}). To help visualize the temperature distribution in the reactor, Fig. 5(d) shows an example plot of Eq. (9) for three adjacent filament wires.

To compare Eq. (9) to experimental data, we first assume that the measured temperature of the stage is approximately equal to the average of the stage and lid temperatures. Justifying this assumption, our data show that changes in stage temperature during experiments affect the overall driving force ($\phi_{\text{stage}} - \phi_{\text{fil}}$) by $\sim 5\%$, meaning that slight differences in T_{stage} and T_{lid} have a minor effect on model validity.

To relate ϕ_R to ϕ_{fil} , and the filament temperature (which we measure), we propose an interfacial heat transfer resistance in accordance with models of the Knudsen jump in planar geometry,²⁰

$$\phi_{\text{fil}} - \phi_R = -\frac{b\lambda q_{\text{fil}}}{2\pi R_{\text{fil}}}. \quad (10)$$

The mean free path is calculated from Eq. (1) using collision diameters of 0.58 and 0.358 nm⁴¹ for IPA and argon, respectively. The diameter of IPA was approximated using Eq. (9-4.8) in Poling⁴⁰ and a critical volume of 222 cm³/mol.⁴¹ The dimensionless interfacial heat transfer resistance b is approximately described by Eq. (11) according to the kinetic theory of gases and the Boltzmann theory of interfacial heat transfer,^{20,33}

$$b = \frac{8}{9} \frac{2 - \alpha_0}{\alpha_0(\gamma + 1)}, \quad (11)$$

where α_0 is the accommodation coefficient or the fraction of inelastic collisions with the filament. Note that in fitting experimental data, the heat capacity ratio was 1.105 for IPA⁵¹ and 1.66 for argon (the value for a monatomic ideal gas). Explanation of how Eqs. (10) and (11) are extracted from Su *et al.*²⁰ can be found in the [supplementary material](#). Combining Eqs. (9) and (10) gives Eq. (12), which models filament temperature based on the overall

heat transfer from the lid to the stage,

$$\phi_{\text{fil}} = \phi_{\text{stage}} - \frac{q_{\text{fil}}}{2\pi} \left(\ln \left(\frac{L}{2R_{\text{fil}}\pi} \right) + \left(\frac{H_1 H_2}{H_1 + H_2} \right) \frac{2\pi}{L} + \frac{b\lambda}{R_{\text{fil}}} \right). \quad (12)$$

Further details on the derivation of Eq. (12) can be found in the [supplementary material](#).

It should be noted that because Eq. (12) explicitly accounts for filament spacing and distance from the stage and distance from the lid, it can be generalized to most reactor geometries relevant to the iCVD study.

E. Heat transfer measurements

Data were collected and processed according to the following procedures. At low pressure, q_{fil} (heat transfer by conduction from the filament array) becomes negligible, so we can solve for the wire emissivity from this limit.¹⁹ To do this, the filament temperature was measured at base pressure (~ 3 mTorr) while varying power to the filament. Temperature versus power data was then fitted to Eq. (3), by varying the wire emissivity, assuming q_{fil} to be zero.

This fit is shown in Fig. 6(a) and corresponds to an emissivity of 2.14, which is notably unphysical (emissivity must be less than 1). It is likely that this high emissivity value is caused by a small leakage of current through the feedthroughs to the reactor, resulting in a mismatch between the current reading from the power supply and the actual current passing through the filament array. Other factors, like error in the length of the filament or conductive losses, are not large enough to cause such a discrepancy. In all calculations, the current is adjusted by a constant factor such that the emissivity of the array is 0.95.

From the known power to the array and radiative heat loss, q_{fil} can be determined from Eq. (3). By collecting power and filament temperature data at a range of chamber pressures (up to 1 Torr) and calculating q_{fil} , we can then fit Eq. (12) to experimental data. Before fitting Eq. (12), however, initial guesses for the fitting coefficients, κ' and α_0 , were determined. This was done by ignoring the temperature dependence of the mean free path (calculating it at a fixed 450 K). For each chamber pressure, this allowed the proportionality constant relating q_{fil} to the thermal driving force ($\phi_{\text{fil}} - \phi_{\text{stage}}$) to be calculated and fit to Eq. (12), to obtain initial values for the fitting parameters κ' and α_0 [Fig. 6(b)]. This proportionality constant can be thought of as overall thermal conductivity, with Fig. 6(b) clearly showing that the reactor vapor phase becomes more insulating as pressure drops for both gases. Note that because of heat losses, filament temperatures were lower at higher pressures, so the nominal 450 K for mean-free path estimation leads to an apparent over-prediction of the filament temperature for the data taken at 1 Torr. Figure 6(b) also clearly shows that there is little pressure dependence above pressures of roughly 400 mTorr and 1 Torr for IPA and argon, respectively. These correspond to a critical Kn of 0.4–0.45. Assuming the same critical Kn, we expect to see very little pressure dependence for monomer and initiator molecules commonly used in iCVD at normal operating pressures.

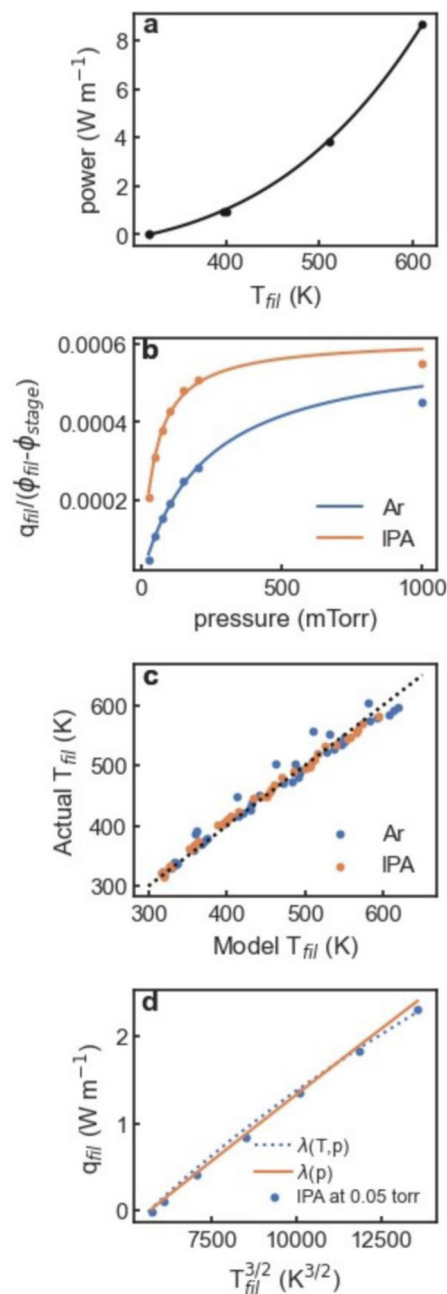


FIG. 6. Experimental data and curve fits of heat transfer from the filament array to the reactor body. (a) Power vs filament temperature in the low-pressure limit (3 mTorr), which fits well to Eq. (3). (b) Overall thermal conductivity vs pressure used to obtain initial values for the fitting parameters in Eq. (12), where temperature dependence of mean free path is neglected. (c) Filament temperature data (blue / dark gray and orange / light gray dots) vs model filament temperature (dotted line) based on Eq. (12) for IPA (orange / light gray) and argon (blue / dark gray). (d) Curve fits for conducted heat flow vs filament temperature, for Eq. (12) when we account for temperature dependence of the mean free path (dotted line) and ignoring the temperature dependence of the mean free path (solid line).

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Using the initial values from the fit shown in Fig. 6(b), the fit was repeated [except this time, the mean free path [Eq. (1)] was evaluated at the filament temperature]. Data were fit to Eq. (12), by varying κ' and α_0 to minimize the mean squared error between the predicted and measured values of the filament temperature. The fitting data encompass the range of pressures and filament temperatures commonly used in iCVD for both IPA vapor and argon gas. The resulting predicted versus measured T_{fil} values are plotted in Fig. 6(c). Fitting parameters from Eq. (12) are included in Table III. To show how the overall thermal conductivity plotted in Fig. 6(b) compares to the fitting results shown in Fig. 6(c), q_{fil} is plotted against filament temperature with the fits based on temperature-independent and temperature-dependent mean free path, respectively, for IPA at 50 mTorr [Fig. 6(d)]. This comparison shows little impact from the temperature dependence of the mean free path.

The thermal conductivity of IPA from Lange's Handbook⁴² is $0.025 \text{ W m}^{-1} \text{ K}^{-1}$ at 250°C , giving a κ' of $0.0011 \text{ W m}^{-1} \text{ K}^{-3/2}$, while argon is reported to have a thermal conductivity of $0.0211 \text{ W m}^{-1} \text{ K}^{-1}$ at 100°C , giving a κ' of $0.00109 \text{ W m}^{-1} \text{ K}^{-3/2}$. The values of κ' reported in Table III match well with the literature values. They vary from the reported values by the same constant, which is likely due to slight inaccuracy in the system parameters or edge effects not accounted for in Eq. (12).

The accommodation coefficient α_0 describes the impact of the Knudsen jump based on a calculation of the mean free path from Eq. (1). The values of α_0 show that when correcting for the molecular diameter in the mean free path, the argon has a less efficient thermal accommodation²⁰ at the surface of the filament. It is important to note that a planar model was used to approximate the filament array when describing the relationship between the Knudsen jump and the accommodation coefficient. The planar model may not apply to the cylindrical geometry of the actual filaments and could contribute to the low α_0 values. Furthermore, the outliers in Fig. 6(c) (the ones not well predicted by the heat transfer model) correspond to argon gas at low pressure. Under these conditions, the assumption of Eq. (10), corresponding to a planar Knudsen jump, is less appropriate, and geometry-dependent corrections⁵² can be introduced if necessary.

Our results for heat transfer from the filament in the iCVD system show that the pressure dependence of conduction can be well accounted for by models of the heat transfer. Furthermore, they show that the same conduction model is valid for gases that absorb infrared radiation (IPA) and those that do not (argon). As such, the predictive ability of our model experimentally validates that infrared absorption can be neglected at iCVD conditions, as was discussed in Sec. III A.

The results from this work have been helpful in our related work elucidating the thermal decomposition of di-*tert*-butyl

peroxide at the filament array.⁴⁷ The model we propose corresponds well with the existing theory of heat transfer and reported heat transfer coefficients (Table III), suggesting it is a strong descriptor of heat transfer in the iCVD reactor system. However, because our description of heat transfer neglects edge effects, it is still an approximate theory, and further work could improve the precision if needed.

F. Reactor effective temperature

When operating the iCVD reactor in batch mode,^{17,27,28,30} the ideal gas law is a helpful model. In fact, most studies of iCVD implicitly use the ideal gas law to perform flow rate calibration before running a deposition. Whenever the ideal gas law is used in these situations, the temperature of the reactor vapor must be specified; however, because the temperature in the reactor is nonuniform, it is not immediately clear what the temperature of the reactor is. The effect of temperature gradients becomes even more pronounced in any batch mode reaction or deposition, such as those that are used to perform condensed droplet polymerization.^{27,28} As batch processes become more common in the literature,^{17,30} the analysis and control of these processes will require a precisely defined temperature. Herein, we refer to the temperature consistent with treating the reactor with the ideal gas law as the effective temperature (T_{eff}). By turning on the filament array while the reactor is closed off from the pump, we measured the change in the effective temperature caused by the filament array.

To define the effective temperature, we begin by assuming (based on Sec. III A) that compositional and pressure gradients in the iCVD reactor are small, but we acknowledge that due to thermal gradients, vapor density ρ varies with location x such that at any location the ideal gas law is valid as described by Eq. (13),

$$T(x) = \frac{p}{R_G \rho(x)}. \quad (13)$$

Integrating Eq. (13) over the volume of the reactor, we obtain Eq. (14),

$$\int \frac{1}{T} dV = \frac{nR_G}{p}, \quad (14)$$

where n is the total number of moles of gas in the reactor. To appeal to the ideal gas law for the reactor overall, we define the effective temperature of the reactor such that the ideal gas law holds for the whole reactor ($pV_r = nR_G T_{\text{eff}}$), giving Eq. (15),

$$T_{\text{eff}} = \frac{V_r}{\int \frac{1}{T} dV_r}, \quad (15)$$

where V_r is the reactor volume. Because it is defined by the ideal gas law, changes in the effective temperature caused by turning on the filament array in an isolated reactor can be measured using chamber pressure since there is a direct proportionality between pressure and temperature for an ideal gas in a closed system. In practice, the reactor is not perfectly closed due to the effect of leak,

TABLE III. Parameters used to fit Eq. (12).

	κ' ($\text{W m}^{-1} \text{ K}^{-3/2}$)	α_0 (nondim)
IPA	0.001 19	0.33
Ar	0.001 17	0.15

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so we instead use a leak-corrected expression for the effective temperature given by Eq. (16),

$$\frac{T_{\text{eff}}}{T_{\text{eff}}^0} = \frac{p}{p^0 + \int \frac{dp}{dt_{\text{leak}}} dt}, \quad (16)$$

where T_{eff}^0 is the effective temperature of the reactor during leak rate measurement, p^0 is the initial pressure in the reactor with the filament off, $\frac{dp}{dt_{\text{leak}}}$ is the rate of pressure change caused by leak into the reactor, and t is the time since the filament was turned on. A detailed derivation of Eq. (16) may be found in the [supplementary material](#).

Equation (16) provides a method for determining changes in T_{eff} from pressure changes in the reactor when the power to the filament array is increased in a stepwise manner. Because the leak rate can change slightly when the temperature of the reactor changes, $\frac{dp}{dt_{\text{leak}}}$ must be remeasured throughout the experiment. Immediately following a step change in the power to the array when the reactor is equilibrating to a new condition, $\frac{dp}{dt_{\text{leak}}}$ cannot be measured but must instead be updated once the reactor has reached a new equilibrium.

To extract a reliable integrated leak rate from chamber pressure and filament temperature data over time, we iterated through the data systematically. We consider the data in 30 s pieces (six data points) by iterating through every 30 s period, i.e., the first period contains data points 1–6, and the second period contains data points 2–7. If the difference between the maximum and minimum filament temperature over the 30 s period is less than 2.7 K, we assume the reactor is equilibrated and update $\frac{dp}{dt_{\text{leak}}}$ as the slope of pressure change over the 30 s period determined from a linear regression. The “window size” of 30 s and temperature cutoff of 2.7 K were both chosen to maximize the number of updates to the leak rate, without analyzing too short a time span or introducing bias from thermal effects. The leak rates determined using this method are subsequently interpolated over all the conditions that contained greater than 2.7 K spread in temperature and were numerically integrated using a Reimann sum to give the total pressure change due to leak since the beginning of the experiment. This integrated leak rate is used in Eq. (16) to determine the effective temperature.

To model the effective temperature obtained from Eq. (16), we focus on the two-chamber design as that is the most common reactor configuration and matches our system. It is useful to consider the bottom and top chambers separately as volumes V_1 and V_2 , respectively, with the former effectively isolated from the filament and the latter heavily affected by the state of the filament array. Alternative models would be straightforward to derive for other reactor designs. While Eq. (9) provides a closed-form approximation for the temperature field surrounding the filaments, it is unnecessarily detailed for determining effective temperature. To obtain a tractable final expression, the filament array is modeled as a planar heat source at the midplane between the stage and lid, yielding a linear temperature profile (neglecting edge effects). By linearizing the equations for heat transfer [Eq. (12)], we obtain

Eq. (17),

$$T_{\text{eff}} = \frac{1}{\frac{1-f}{T_{\text{ref}}} + \frac{f \ln \left(1 + h_{\text{eff}} \left(\frac{T_{\text{fil}}}{T_{\text{av}}} - 1 \right) \right)}{h_{\text{eff}} (T_{\text{fil}} - T_{\text{av}})}}, \quad (17)$$

where $f = \frac{V_2}{V_1 + V_2}$ is the fraction of the total reactor volume that can be accurately modeled using Eq. (9), T_{ref} is the reference temperature, which is assumed to be 40 °C based on setpoints for reactor heating, and h_{eff} is defined by Eq. (18),

$$h_{\text{eff}} = \frac{\left(\frac{H_1 H_2}{H_1 + H_2} \right) \frac{2\pi}{L}}{\ln \left(\frac{L}{2R_{\text{fil}}\pi} \right) + \left(\frac{H_1 H_2}{H_1 + H_2} \right) \frac{2\pi}{L} + \frac{b\lambda}{R_{\text{fil}}}}. \quad (18)$$

Details of the derivation of Eqs. (17) and (18) can be found in the [supplementary material](#).

To fit the effective temperature to Eq. (17), the fraction f was calculated from the area of the filament array (371 cm²) multiplied by the height of the chamber divided by the total reactor volume. This gave a value of 0.32 for f . The distance between filament wires, chamber height, and radius of the filament wire correspond to those used in Sec. III D. Equation (17) was fit to experimental data [from Eq. (16)] by minimizing the mean squared error.

The fitting coefficient α_0 is reported in Table IV. The accommodation coefficient α_0 from Table IV corresponds excellently with the same value found from the heat transfer analysis in Table III for argon. However, the value obtained for IPA is markedly lower. Alone this does not suggest a major source of error in the model since the effective temperature is not strongly impacted by the value of the accommodation coefficient.

Curve fits in argon [Figs. 7(a)–7(d)] and IPA [Figs. 8(a)–8(d)] (gases that do not decompose at high temperature) show that Eq. (17) is an acceptable representation of the effective temperature of the vapor phase. Importantly, over a large range of filament temperature values, the effective temperature changes by only ~5%–10%. The notably higher noise in low-pressure data [Figs. 7(a) and 8(a)] is caused by the increased effect of leak at low pressure. Overall, the effect of pressure change is minimal. There is some notable deviation from the shape of the model and the shape of the data at temperatures above ~550 K for both IPA (Fig. 8) and, to a lesser degree, argon (Fig. 7). Because this anomaly is observed in argon gas, which does not absorb infrared radiation, it is likely caused by increased leak or radiative heating

TABLE IV. Accommodation coefficient for argon and IPA fit from effective temperature.

	α_0 (nondim)
IPA	0.064
Argon	0.16

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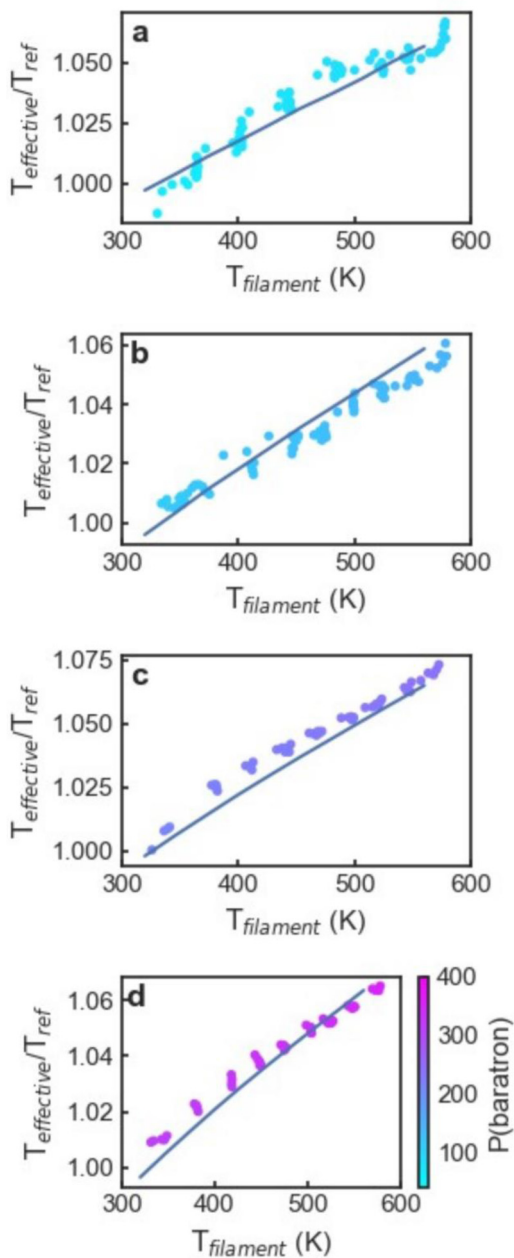


FIG. 7. Ratio of effective temperature to reference temperature vs filament temperature for argon at nominal pressures of (a) 50, (b) 100, (c) 200, and (d) 300 mTorr.

of the reactor body, which becomes much more pronounced at high filament temperatures due to the quartic scaling of radiative power with filament temperature [Eq. (3)]. Accounting for radiation is outside the scope of work as such high filament temperatures are rarely used.

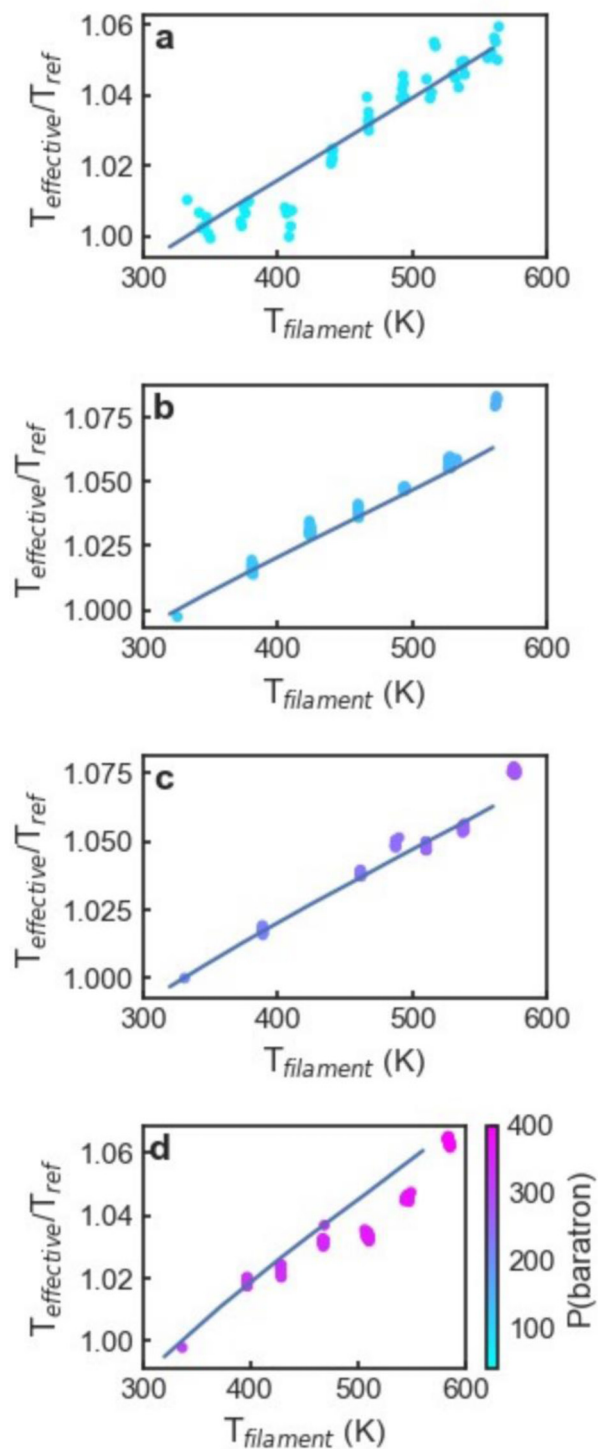


FIG. 8. Ratio of effective temperature to reference temperature vs filament temperature for IPA at nominal pressures of (a) 50, (b) 100, (c) 200, and (d) 300 mTorr.

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IV. SUMMARY AND CONCLUSIONS

The published iCVD literature contains very little discussion of the fundamental processes underlying vapor-phase mixing and heat transfer in the reactor. This study attempts to clarify the fundamental processes that underly contacting patterns in the vapor phase, heat transfer from the filament, and heat transfer from the reactor body. We present an overview of the dimensionless numbers important to these three cases. Detailed analysis of Pe_m in mass transfer and Kn in heat transfer from the array has been considered. This study enables more precise measurements from the iCVD reactor, a concrete understanding of the conditions under which film uniformity or kinetics may be affected by poor mixing, and the temperature profile and heat transfer from the filament array. These capabilities are necessary to study the initiation process, which has largely gone unexplained¹³ and is the subject of our recent work.⁴⁷

For mass transfer, we find that the iCVD reactor is ideally well mixed, and that deviations from perfect mixing can be fully quantified by Pe_m . We validate the perspective that under ideal conditions, the reactor is well mixed by performing step-response experiments with IPA and argon using a Pirani gauge for real-time composition sensing. We find that the resulting step-response curve matches the design equation for a well-mixed reactor.²¹ To visualize the role of mixing in the reactor, we performed Autodesk CFD simulations at a range of Pe_m to replicate the step-response experiments. These show that outlet gas composition provides poor sensitivity to mixing, whereas comparing readings from two sensors placed at disparate reactor locations gives strong sensitivity to nonideal mixing.

We also performed simulations of binary gas mixing at the same range of Pe_m with both a one-chamber and two-chamber reactor design. These simulations demonstrated that the mixing was not impacted by the additional chamber, illustrating the utility of simulation in design innovation. Furthermore, these simulations illustrate the potential impact of poor mixing on deposition over the full range of transitional Pe_m . Our simulations suggest that below Pe_m of 0.3, the reactor can be considered as well-mixed. Outside the well-mixed regime, film nonuniformities may arise. Additionally, our work illustrates the potential of the simple, low-cost, and precise Pirani gauge for characterizing vapor composition.

For heat transfer, analysis of dimensionless numbers shows that ideal heat transfer to the process gas occurs at low Kn via conduction with no impact from the Knudsen jump. Interfacial heat transfer resistances developing at low pressures can be quantified by Kn . To validate this perspective of ideal heat transfer, we build on past work from Bakker *et al.*¹⁹ to measure heat transfer from the filament wires. Our data agree with an approximation of the Knudsen jump based on the planar geometry even though the filament is cylindrical.²⁰ However, the use of a planar geometry likely causes error for low-pressure argon. Temperature-independent thermal conductivities of 0.00119 and $0.00117 \text{ W m}^{-1} \text{ K}^{-3/2}$ were obtained for IPA and argon, respectively, compared with values of 0.0011 and $0.00109 \text{ W m}^{-1} \text{ K}^{-3/2}$ from the literature.⁴² Both measured values correspond well with the literature and differ from the literature by a constant factor, indicating that there may be some

impact from edge effects or inaccurate dimensions used in our model. While our heat transfer model makes significant approximations, its fit to data and correspondence with gas thermal conductivity reported in the literature suggests that it captures the primary physical drivers of heat transfer to the process vapor. For Kn greater than 0.4, heat transfer nonidealities are expected to appear.

While it may appear from our data that heat transfer is strongly affected by pressure at normal iCVD operating pressures, it should be noted that IPA and argon have much smaller collision diameters than normal iCVD process gases; therefore, the Knudsen jump is often negligible under normal iCVD conditions.¹⁹ The heat transfer model presented provides a fundamental understanding of the heat transfer process from the filament arrays, enabling future insight into methods of improving the initiation efficiency during iCVD. The correspondence between predicted thermal conductivity from our model with the literature further supports the treatment of neglecting absorption of thermal radiation by the process vapor and thermal convection.

To make the application of the ideal gas law rigorous in the iCVD reactor system, we introduce the concept of an effective reactor temperature and provide a model for it. By turning on the filament array while it is closed off from the pump, we measured the change in the effective temperature caused by the filament array, finding a strong model prediction and an overall change in the effective temperature of only 5%–10%. Notably, fitting the effective temperature yielded accommodation coefficients of 0.06 and 0.16 for IPA and argon, respectively. Because the effective temperature depends weakly on the accommodation coefficient, there is high error in the accommodation coefficient predicted by effective temperature. While the accommodation coefficient values obtained from the measurement of diffusive heat transfer in argon (0.15) matched well, that for IPA (0.33) was significantly higher. While this difference could be caused by absorption of thermal radiation, further experiments are needed to accurately test such a hypothesis.

Given the diversity of innovations originating from efforts to push the boundaries of deposition at high monomer saturation,^{26–29} we hope that our work inspires similar innovation at high Pe_m or Kn . Because the step-response experiments performed here used IPA and argon, lower diffusivity systems that better represent iCVD process gases should be studied to more thoroughly test the predictions made in Sec. III B. We have left experimental investigation of Knudsen flow (high Kn), thermal convection (high Pe_t), and absorption of thermal radiation by the process gas to future work. We also note that further investigations are needed to explain the apparent increase in effective temperature at high filament temperature. Additionally, innovative reactor design is needed to obtain uniform films from iCVD at high Pe_m . For these investigations, more extensive and complete reactor simulations would be eminently useful. Having introduced the Pirani gauge as a thermal conductivity-based composition sensor, correlations between the Pirani gauge reading and the Knudsen jump phenomenon at the filament may also prove useful. Finally, it is important to apply the fundamental understanding gained from these experiments to advance the understanding of chemical reactions in iCVD, using the temperature profile and well-mixed

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design equation for computational modeling of vapor-phase reactions.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for names of animation files for step-response; details of residence time distribution model including additional data for $Pe = 0.03$ simulation and the calculation of reactor volume from the residence time distribution; derivation of expressions relating heat transfer variables to literature values of accommodation coefficient and thermal conductivity [Eqs. (10) and (11) from the main text]; derivation of the model used for heat transfer from the filament; derivation of the model used for the effective temperature, including the definition of the effective temperature; and derivation of the equation for measuring effective temperature and the reactor effective temperature model.

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AUTHOR DECLARATIONS

Conflict of Interest

R.Y. has financial interests in Oto Biotech Inc. R.Y.'s interests are reviewed and managed by Cornell University in accordance with their conflict-of-interest policies.

Author Contributions

Simon Shindler: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Funding acquisition (lead); Investigation (lead); Methodology (lead); Project administration (lead); Resources (lead); Supervision (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (supporting). **Rong Yang:** Conceptualization (equal); Data curation (lead); Formal analysis (supporting); Funding acquisition (lead); Investigation (supporting); Methodology (supporting); Project administration (lead); Resources (lead); Supervision (lead); Validation (lead); Visualization (lead); Writing – original draft (supporting); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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